AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM112
Introduction to Computing and Information Technology

Final Examination
Summer Semester 2025/2026

Time Allowed: 3 Hours

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Instructions:

- Total Marks: ....
- This exam consists of ... parts.
- ALL questions must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: Multiple Choice Questions (20 marks).

1. Which cloud deployment model combines public and private clouds?
   a) Public Cloud
   b) Community Cloud
   c) Hybrid Cloud
   d) Distributed Cloud
2. Which cloud service allows customers to develop and deploy applications without managing the underlying hardware?
   a) SaaS
   b) PaaS
   c) IaaS
   d) Hybrid Cloud
3. Which smartphone component converts analog signals into digital signals?
   a) GPU
   b) CPU
   c) ADC (Analog-to-Digital Converter)
   d) DSP
4. Which operating system component remains in memory while the computer is running?
   a) BIOS
   b) Kernel
   c) Cache
   d) Bootloader
5. Which memory loses its contents when power is turned off?
   a) ROM
   b) SSD
   c) RAM
   d) Cache
6. Which malware can self-replicate through networks?
   a) Trojan
   b) Worm
   c) Spyware
   d) Adware
7. Which attack tries every possible password combination?
   a) Dictionary Attack
   b) Brute-force Attack
   c) Phishing
   d) Spoofing
8. Which authentication factor is represented by a fingerprint?
   a) Something you know
   b) Something you have
   c) Something you are
   d) Something you own
9. What does IoT stand for?
   a) Internet of Technology
   b) Internet of Things
   c) Internal Operating Tool
   d) Internet of Telecom
10. Bluetooth usually works within:
    a) 1 km
    b) 500 m
    c) Less than 10 m
    d) 100 km

Part 2: Short questions (30 marks)

Question 1: (6 marks)
Define the following:
   a) Cloud Computing
   b) Botnet
   c) Two-Factor Authentication

Question 2: (6 marks)
Differentiate between:
a) IaaS and SaaS
b) RAM and ROM

Question 3: (6 marks)
Name three types of malware.

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Question 4: [6 marks]
State two advantages of multiprocessor systems.

Question 5: [6 marks]
Differentiate between:
a) Authentication and Authorization
b) Bluetooth and Wi-Fi Direct

Part 3: Problem Solving / Long Questions (50 marks)

Question 1: (10 marks)
A company stores employee passwords as plain text.
a) Explain why this is dangerous.
b) Suggest a better solution.

Question 2: (10 marks)
A company wants to provide smartphones to all employees working remotely.
a) Explain the importance of the following smartphone components:
- CPU
- RAM
- Flash Memory
- RF Module
- Battery

b) Discuss how these components affect the performance of the device.

Question 3: [10 marks]
Two users are running many applications simultaneously on the same computer.
a) Explain how the operating system manages multiple programs at the same time.
b) Describe the roles of:
- Process Management
- Memory Management
- CPU Scheduling

c) Explain why multitasking improves user experience.

Question 4: [10 marks]
A user receives an email asking them to reset their bank password through a suspicious website.
a) Identify the attack.
b) Explain two risks.
c) Suggest four security practices.

Question 5: [10 marks]
a) Explain how Bluetooth differs from NFC.
b) Explain three challenges facing mobile devices.

End of questions

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